Hastings Technology Metals

ANNUAL REPORT

For the Period

1 December 2017 to 30 November 2018

COMBINED GROUP REPORT C265/2008

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Attachment 2 **PETROGRAPHIC REPORT 14 Samples from Drillholes AUDD0178, AUDD0182 and AURC198 from the Auer Prospect**
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Attachment 3 **Yangibana Project Mineral Resource Estimate October 2018**
Hastings Yangibana Resource Estimation Report October 2018 - V2.docx

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HAS_RX_20190130.jpg

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Attachment 3 **Drilling Attachments**
Drilling.zip
ASSAY_COLLAR_LITH_SURVEY.zip

ATTACHMENTS SUBMITTED SEPARATELY

1. Bibliographic Data Sheet

Project Name:	Yangibana Rare Earth's Project
Combined Reporting Number:	
Tenement Numbers:	E 09/01043,E 09/01700,E 09/01703,E 09/01704,E 09/01705,E 09/01706,E 09/01943,E 09/01944,E 09/01989,E 09/02007,E 09/02018,E 09/02084,E 09/02086,E 09/02095,E 09/02129,E 09/02137,P 09/00482,P 09/00489,M 09/00157,M 09/00158,M 09/00159,M 09/00160,M 09/00161,M 09/00162,M 09/00163,M 09/00164,M 09/00165
Tenement Operator(s):	Hastings Technology Metals
Report Type:	Annual
Report Title:	COMBINED GROUP REPORT C265/2008
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Map Sheets:	*1:250,000 Map Sheet*	*1:100,000 Map Sheet*
	SF50-14 (EDMUND)	2150 (EDMUND)
	SG50-02 (MOUNT PHILLIPS)	2149 (MOUNT PHILLIPS)

Target Commodity:	RARE EARTH ELEMENTS
Prospects Drilled:	Bald Hill, Frasers, Auer North, Auer, Yangibana
PoW Number:	72272, 60478, 66332
Geophysical Survey Reg No:	
Assays:	

Abstract

Location: The Yangibana Project is located 270 kilometres east-northeast of Carnarvon, within the Upper Gascoyne Shire of Western Australia.

Geology: The project area is located within the Gascoyne Province. Local geology predominantly consists of the Yangibana and Pimbyana Granites which form part of the Durlacher Supersuite.
Within the project area these granites have been intruded by a number of dykes and sills of carbonatitic affinity with associated fenitic halos. Numerous rareearths-bearing ironstone veins are also present within the project area, which are the focus of exploration activities.

Work Done: A total of 372 drillholes were completed during the reporting period, for a total of 20,126.05m. This drilling was completed at Bald Hill, Fraser's, Auer, Auer North, Yangibana North, and Yangibana.

Results: Exploration completed during 2018 was successful in converting resources to reserves in the pre-mentioned prospects.

Conclusion: Further drilling is planned for 2019, in particular the T1 magnetic target identified through geophysical processing during 2018 will be tested.

Drilling Summary:

Hole Type	No. of Holes	Total Drilled (m)
DD	41	2969
RC	331	17157

Surface Geochemistry Summary:

Sample Type	No. of Samples
Rock chip	43

2. Introduction

During 2018 a total of 372 drillholes were completed on the Yangibana Project, for a total of 20,126.05m, including 2969.05m of diamond drilling. Drilling at Bald Hill and Fraser's was predominantly designed to provide sample for a pilot plant programme, with drilling at Auer and Auer North completed for resource definition and expansion.

Geotechnical drilling (DD) was carried out in Yangibana, Auer and Auer North. 19 geotechnical holes were drilled for a total of 1887.4m which is included in the total metres quoted above. This drilling was completed for ongoing mining studies to allow conversion of resources to reserves.

Following completion of drilling the Mineral Resource Estimate for the project was updated, and now stands at 21.67 Mt @ 1.17% TREO including 0.40% Nd_2O_3 Pr_6O_{11}.

Further geophysical modelling and processing was completed utilising the airborne magnetic data acquired during 2016. This work focused on extensions to Auer and Auer North, and surrounding areas. Inversion modelling of the southern extension to Auer led to successful drill testing.

The geophysical modelling also identified a large, deep target in the southeast of the project area. This target, currently termed T1, is scheduled for drill testing in 2019.

Metallurgical work continued throughout the year on various samples.

An Exploration Index Plan is provided as Figure 1.

3. Location and Access Details

The Yangibana Project is located approximately 270 kilometres east northeast of Carnarvon, within the Upper Gascoyne Shire of Western Australia (Figure 2). The project is predominantly located within the Edmund 1:250,000 map sheet, with the southern portion of the project area located upon the Mt Phillips 1:250,000 map sheet.
Access is by sealed road from Carnarvon to Gascoyne Junction, and then by gravel road from Gascoyne Junction to Gifford Creek Homestead. Numerous station tracks provide access within the project area.

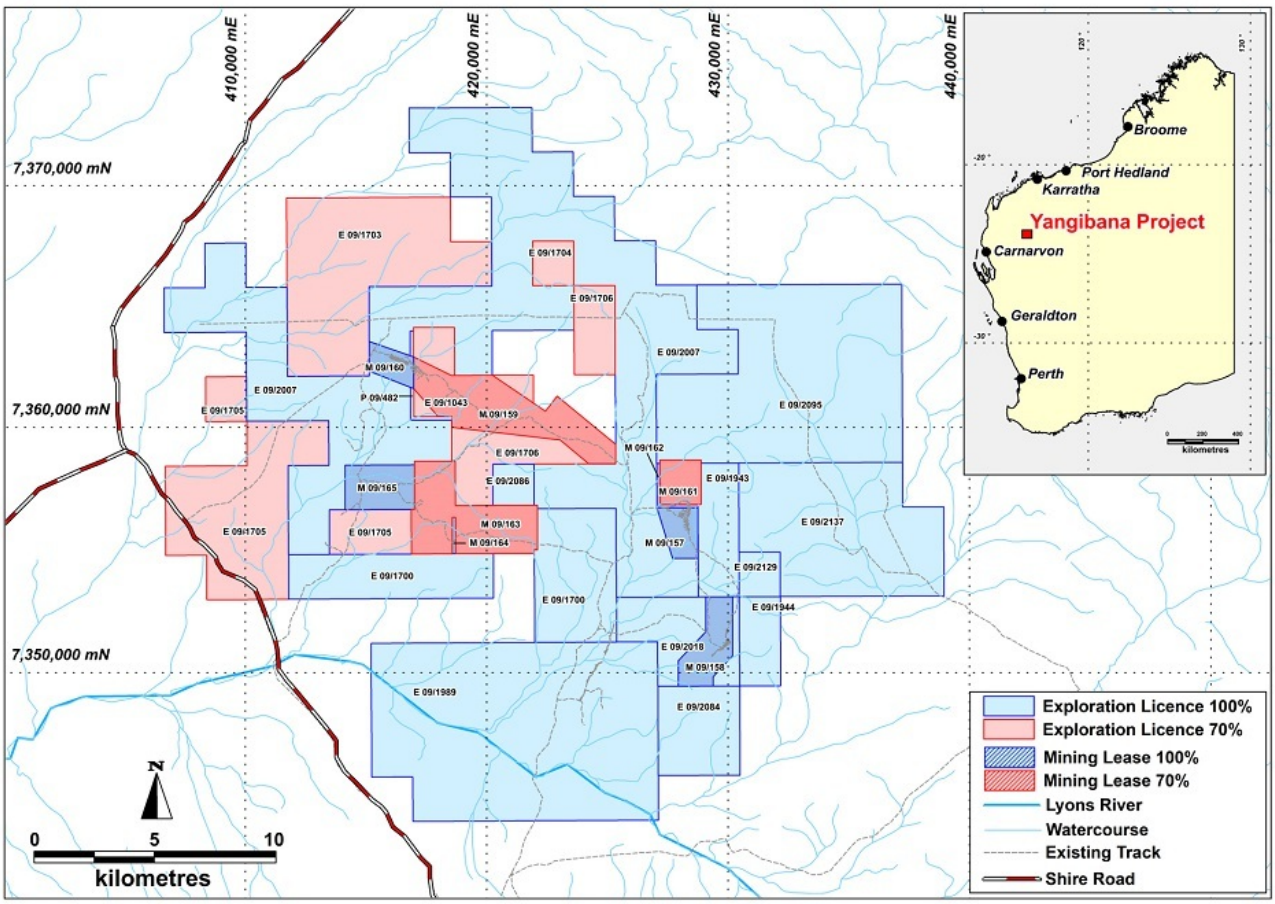


Figure 2: Yangibana Project Combined Reporting Group C265/2008 Location and Tenure

4. Tenement Details

Tenement Information

Tenement	Grant Date	Expiry Date	Holder	Expenditure ($)	Area Size (KM2)	Area Size (BLK)
E 09/1043	01/12/2004	30/11/2018	GASCOYNE METALS PTY LTD,MOJITO RESOURCES LIMITED	20000	2.8	1
E 09/1700	20/05/2011	19/05/2021	YANGIBANA PTY LTD	70000	30.8	11
E 09/1703	01/04/2011	31/03/2021	GASCOYNE METALS PTY LTD,MOJITO RESOURCES LIMITED	70000	36.4	13
E 09/1704	01/04/2011	31/03/2021	GASCOYNE METALS PTY LTD,MOJITO RESOURCES LIMITED	20000	2.8	1
E 09/1705	01/04/2011	31/03/2021	GASCOYNE METALS PTY LTD,MOJITO RESOURCES LIMITED	70000	36.4	13
E 09/1706	01/04/2011	31/03/2021	GASCOYNE METALS PTY LTD,MOJITO RESOURCES LIMITED	70000	22.4	8
E 09/1943	31/12/2012	30/12/2022	YANGIBANA PTY LTD	30000	8.4	3
E 09/1944	31/12/2012	30/12/2022	YANGIBANA PTY LTD	30000	5.6	2
E 09/1989	13/06/2014	12/06/2019	GASCOYNE METALS PTY LTD	40500	75.6	27
E 09/2007	27/11/2013	26/11/2023	GASCOYNE METALS PTY LTD	96000	134.4	48
E 09/2018	28/06/2013	27/06/2023	YANGIBANA PTY LTD	30000	14	5
E 09/2084	31/10/2014	30/10/2019	GASCOYNE METALS PTY LTD	20000	11.2	4
E 09/2086	31/10/2014	30/10/2019	GASCOYNE METALS PTY LTD	10000	2.8	1
E 09/2095	18/11/2014	17/11/2019	GASCOYNE METALS PTY LTD	31500	58.8	21
E 09/2129	12/06/2015	11/06/2020	GASCOYNE METALS PTY LTD	10000	2.8	1
E 09/2137	07/09/2015	06/09/2020	GASCOYNE METALS PTY LTD	30000	36.4	13
P 09/482	07/10/2014	06/10/2022	GASCOYNE METALS PTY LTD	2000	0.41	0
P 09/489	08/12/2015	07/12/2019	GASCOYNE METALS PTY LTD	2000	0.14	0
M 09/157	01/07/2015	30/06/2036	GASCOYNE METALS PTY LTD	29000	2.9	0
M 09/158	01/07/2015	30/06/2036	YANGIBANA PTY LTD	54000	5.39	0
M 09/159	01/07/2015	30/06/2036	GASCOYNE METALS PTY LTD,MOJITO RESOURCES LIMITED	148000	14.8	0
M 09/160	17/11/2015	16/11/2036	GASCOYNE METALS PTY LTD	23500	2.34	0
M 09/161	25/02/2016	24/02/2037	GASCOYNE METALS PTY LTD,MOJITO RESOURCES LIMITED	31400	3.13	0
M 09/162	25/02/2016	24/02/2037	YANGIBANA PTY LTD	10000	0.48	0
M 09/163	25/02/2016	24/02/2037	GASCOYNE METALS PTY LTD,MOJITO RESOURCES LIMITED	133000	13.3	0
M 09/164	25/02/2016	24/02/2037	GASCOYNE METALS PTY LTD	10000	0.21	0
M 09/165	25/02/2016	24/02/2037	GASCOYNE METALS PTY LTD	53400	5.34	0

The Yangibana Project currently comprises 18 granted exploration licences, 2 granted prospecting licences, and 9 granted mining leases, all held within combined reporting group C265/2008. Table 1 summarises the tenements held within C265/2008. These tenements cover a total area of 647 square kilometres and are shown in Figure 2.
During 2016 a number of areas within existing tenements were converted to mining leases, including:

- A portion of E09/1049 being converted to M09/161;
- A portion of P09/467 being converted to M09/162;
- Portions of E09/1043, E09/1049, and E09/1705 being converted to M09/163;
- A portion of P09/481 being converted to M09/164;
- A portion of E09/2007 being converted to M09/165.

5. Geology

5.1 Regional Geology

The Yangibana Project covers a majority of the Gifford Creek Ferrocarbonatite Suite (GCFS; Johnson, 2013). The GCFS is located within the Gascoyne Province of the Capricorn Orogen, between the Archaean Yilgarn Craton to the south, and the Archaean Pilbara Craton to the north. The Gascoyne Province, which consists of a suite of Archaean to Proterozoic gneisses, granitic and metasedimentary rocks (Johnson et al, 2011), is overlain by the Paleoproterozoic Ashburton Formation to the north, the Mesoproterozoic Edmund and Collier Basins to the east, and the Phanerozoic Carnarvon Basin to the west.

5.2 Local Geology

Locally the tenement area consists of granitic rocks of the 1680-1620 Ma Durlacher Supersuite, specifically the Pimbyana and Yangibana Granites, which contain abundant rafts of sedimentary rocks including arenites, calc-silicates, and quartzites of the 1760-1680 Ma Pooranoo Metamorphics. These rocks are intruded by sills, dykes, and veins of ferrocarbonatite and ironstone, and associated fenitic alteration, which comprise the 1,300-1,280 Ma GCFS (Figure 3; Johnson, 2013).
The area has been subject to several distinct tectonothermal cycles of which the Mangaroon Orogeny (1680-1620Ma), the Mutherbukin Tectonic Event (1385-1200Ma) and the Edmundian Orogen (1030-955Ma) are the most significant. During the Mangaroon Orogeny, intense deformation and metamorphism were coincident with the intrusion of the Durlacher Supersuite (Wallace and Mortimer, 2016; Slezak, 2017). The ferrocarbonatites that occur within the project area have been dated at 1262-986Ma, which are coincident with both the Mutherbukin Tectonic Event and the Edmundian Orogeny. The Lyons River Fault, which is a deep crustal lineament that has been traced to the Mohorovicic discontinuity, may have acted as a fluid conduit during these events, resulting in the emplacement of the ferrocarbonatites and associated hydrothermal alteration over a period of several hundred million years (Slezak, 2017).
The GCFS is dominated by two major southeast trending shear zones, which are cross-cut by multiple north-south to north-east trending faults. The Northern Shear Zone, the Bald Hill Lineament, is anastomosing and variable in width, comprising multiple fault splays (Wallace and Mortimer, 2016). Geophysical interpretation by Southern Geoscience Consultants during 2016 indicated that the Bald Hill Lineament separates an early, highly fractured granite to the north from folded Pooranoo Metamorphics to the south.
The southern shear zone, the Lyons River Fault, is less complex and narrow, forming the boundary between the Edmund Group, and an early granite to the north. A third major shear zone is evident in the geophysical data trending east-west to east-southeast (Wallace and Mortimer, 2016).
The Pooranoo Metamorphics are multiple folded along southeast trending axes. In the northwest these folds are generally wide with granite cores. This area of wider folding has also been intruded by a later granite. In general, the metamorphics are more highly deformed toward the east (Wallace and Mortimer, 2016).

One of the key features of the GCFS within the project area is the widespread occurrence of ironstone dykes that are spatially associated with the ferrocarbonatite intrusions. These are surrounded by relatively narrow haloes of fenitic alteration, and locally associated with quartz veining (Johnson, 2013). Fenitic alteration haloes are characterised by the presence of feldspars and/or Na-amphiboles and magnetite. The ironstone dykes, with dominant north, northeast and west-northwest trends, consist predominantly of goethite, hematite, and magnetite, and are locally weakly radioactive. The ferrocarbonatites and ironstones of the GCFS occur as sinuous pods and veins generally less than 10m wide, that are traceable for up to 25 kilometres. The ironstones of the GCFS have historically been the focus of exploration activity in the area, due to their enrichment in Rare Earths Elements (REE).
The main ore mineral at Yangibana is monazite. The monazite occurs in the ironstone dykes, and are considered to be the weathered remnants of the ferrocarbonatite intrusions and related rock types. Two distinct types of carbonatite have been identified within the project area as described by Slezak, 2017:

1. Fine grained dolomitic carbonatites predominantly in the south of the project area near the Lyons River Fault
2. Macro-crystalline ferrocarbonatites found at depth proximal to ironstones that occur sub-parallel to the Lyons River Fault

The fine grained dolomitic carbonatites are composed predominantly of dolomite, aegirine, riebeckite, magnetite, and apatite with lesser barite and monazite. The macro-crystalline ferrocarbonatites are associated with the ore bearing ironstones. They are composed of coarse ankerite and subordinate calcite, with inclusions and veins of biotite, magnetite, monazite, apatite, and minor potassium feldspar. Phoscorite has also been identified associated with the ore-bearing ironstones, composed of biotite, magnetite, apatite, and monazite (Slezak, 2017).
The potential for the presence of carbonatite and associated intrusions to be present at depth is supported by petrological analysis of samples from both outcrop and drilling which has identified both phoscorite and carbonatite as hosts to mineralisation. This work has indicated that the outcropping ironstones throughout the Yangibana project area may in fact be the weathered and leached remnants of phoscorite intrusions associated with the GCFS. Mineralisation hosted by ferroan dolomite or ankeritic carbonate as identified in drilling, may have exsolved from the phoscoritic parent magma (Crawford, 2016). Identification of these lithologies suggests that there is good potential for further primary mineralisation to be identified at depth below the current level of drilling associated with the phoscorite and/or carbonatite units.
REE mineralisation at Yangibana is predominantly hosted by monazite, a phosphate mineral. Initial petrological assessment of samples of mineralised carbonatite from Yangibana North, identified the monazite as being the Ce rich endmember monazite-(Ce). With the acquisition of further data from other areas within the project area, significant variations in the Nd content from prospect to prospect were identified. Prospects in the Eastern Belt of mineralisation (Bald Hill, Fraser's, Auer, and Auer North) were identified as having significantly higher Nd content than prospects within the Western Belt of mineralisation (Yangibana to Kane's Gossan) suggesting that monazite may vary throughout the project area from the monazite-(Ce) variant, toward more Nd rich variants (monazite-(Nd)).

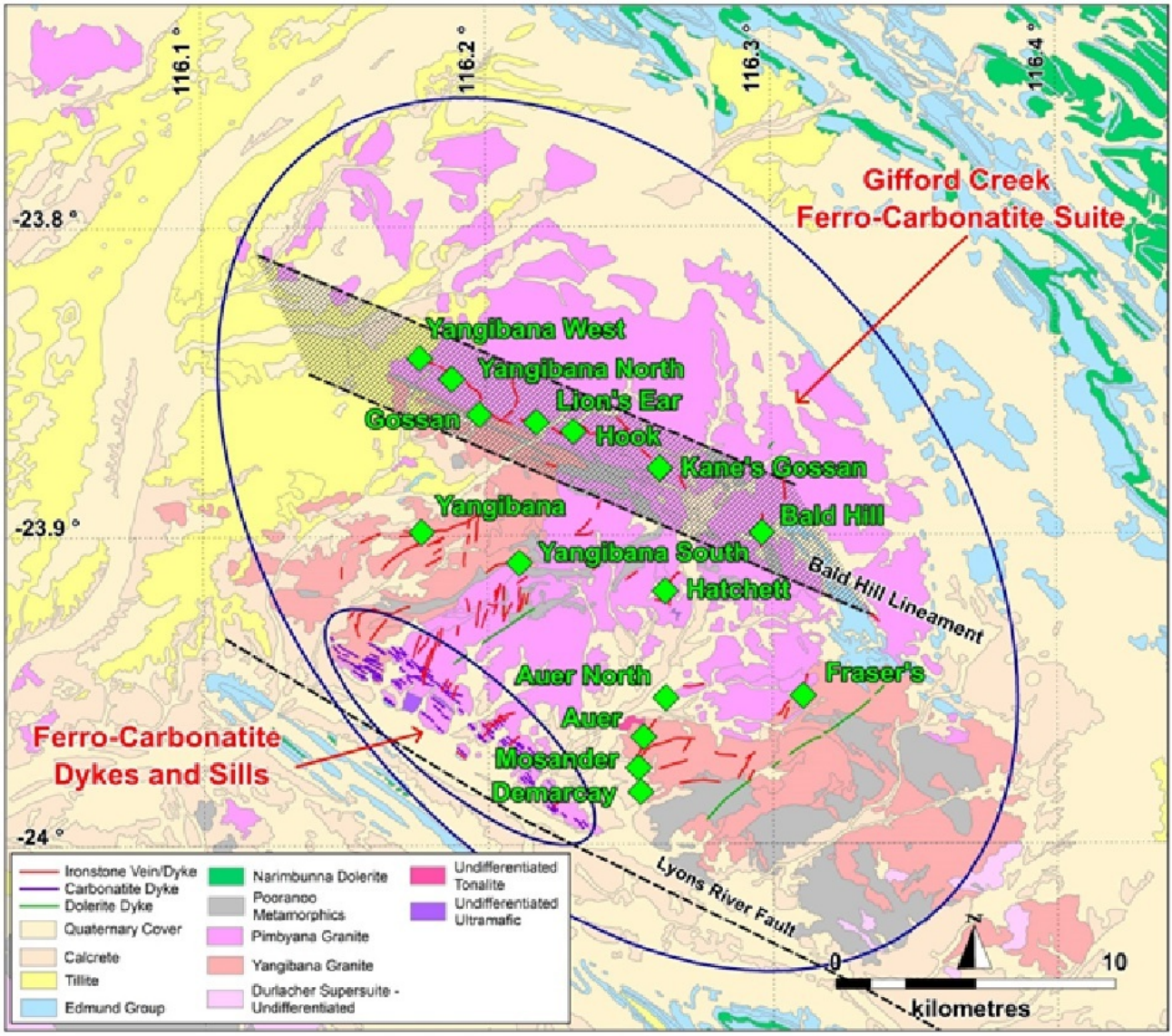


Figure 3: Yangibana Project 1:100,000 Regional Geology and Prospects

6. Previous Exploration

The Yangibana ironstones were first pegged by a prospecting syndicate in 1972, as base metal-bearing banded ironstone targets.
Recorded uranium exploration commenced in 1973 when Noranda Australia Limited (Noranda) investigated an ironstone outcrop with an elevated total count radiometric response (the Yangibana or Noranda Gossan) identified through ground scintillometer traversing and surface geochemical sampling. Maximum uranium values of 53ppm U for an ironstone and 71ppm U for feldspathic wallrock were recorded. As Noranda decided that the radiometric anomalism was largely due to thorium they undertook no further exploration.
From 1974 to 1975 Newmont Pty Ltd (Newmont) explored the area for base metals and focused on five prospects within their tenements and the surrounding area. Work consisted of rock chip sampling and diamond drilling. Results from rock chip sampling identified lead-zinc rich gossans of up to 3% Pb and 1% Zn, with uranium levels of up to 58ppm U at Yangibana East (Noranda Gossan), South Yangibana (Yangibana South), and East Fraser's Well (Fraser's) prospects. Diamond drilling at the South Yangibana and East Fraser's Well prospects intersected base metal mineralisation of up to 0.65% Zn, 0.55% Pb, and 5g/t Ag hosted by a carbonate-rich magnetite-muscovite schist. Analyses for the diamond drilling programme did not include uranium.
In 1981 Cliffs International Incorporated first examined the area for manganese before subsequent photogeological studies identified additional targets for base metals and uranium. One report indicates that 63 rock chip samples were collected and resulting assays recorded anomalous lead, zinc and REE. No records of this sampling have been identified.
Between 1985 and 1991 Challenger Mining Corporation NL and Hurlston Pty Ltd (Challenger; Hurlston) carried out rock chip sampling, mapping, petrology, ground radiometrics and reverse circulation (RC) drilling. In the 1987 Annual Report for E09/95, Challenger reported secondary uranium mineralisation associated with ironstone veins. Surface samples from five main prospects returned up to 1.41% U within ironstone, and up to 0.16% U in granitic wall rocks. REE values were considered to be highly anomalous with up to 1.68% Ce, 0.77% La, and 0.16% Nd reported. Details of analytical methods or laboratories, and exact sample locations have not been identified.
In 1988 Challenger drilled a total of 80 RC drillholes across ten REE prospects along a total strike length of seven kilometres. Challenger calculated average grade and bulk mineralisation tonnage for each prospect which was then totalled into an 'indicated shallow resource potential' of over 3.5 million tonnes at 1.7% rare earth oxide (pre-JORC estimate). It was acknowledged that further drilling was required, however this was never undertaken.
Newmont (later Newcrest Mining NL) resumed exploration of the area in 1989, in a joint venture with Western Rare Earths NL. Although no further prospects were identified, Newcrest recognised the potential of the Gifford Creek Alkaline Complex and initiated a joint research programme with the Key Centre for Strategic Mineral Deposits, University of WA, which culminated in a PhD thesis published in 1996 (Pearson, 1996).
In 2007, Vistarise Limited reviewed the available geophysical data and historic exploration, and concluded that the area warranted modern multi-commodity exploration.
During 2008, GTI Resources Ltd (GTI) completed an open file data compilation and a 56-sample rock chip sampling program over selected sites. Multi-client airborne magnetic and radiometric survey data was also reprocessed and reinterpreted. The overall arithmetic average of the 56 samples collected by GTI was 2.84% TREO. Results from this rock chip sampling programme confirmed REE mineralisation at selected locations. Elevated uranium content previously reported to be associated with ironstones and granitic wall rock by previous explorers was not confirmed.
In 2010, Artemis Resources Ltd (Artemis) undertook a rock and soil sampling program targeting the known east-west striking ironstone lenses. The soil samples were collected from below the last indication of plant matter, generally at approximately 20-40cm depth, with the rock samples taken at the same location. The programme consisted of a series of north trending transects, each 50 metres in length, with a sample collected every 5 metres along each transect. 41 samples of rock and 72 samples of soil were collected. Transects were located at the beginning, end, and middle of each targeted zone. A total of 10 transects were completed across four prospects comprising Yangibana North, Lion's Ear, Hook, and immediately to the west of Hook.
During April of 2010 REE specialist Dr Tony Mariano accompanied Artemis personnel on a site visit. Dr Mariano acquired rock samples for scientific studies and completed a report on the economic potential of the Yangibana Project. In July 2010 Artemis completed an auger drilling programme comprising the collection of 382 samples. The sampling program incorporated both rock and soil at Yangibana. The best results reported were 1.32% Ce, 0.58% La, 0.39% Nd, 0.12% Pr, 649 ppm Sm, and 759ppm Th.

During September 2011 Hastings Rare Metals Limited undertook an introductory site visit to the Yangibana Project. The trip successfully accessed the western portion of the known ironstone outcrop between Yangibana North in the west and Lion's Ear in the east, and also included the southern prospects comprising Tongue, Yangibana and Yangibana South. Thirty-eight grab samples were collected during the visit and were analysed by Genalysis Perth, with assays being in line with expectations based on previous exploration. The most consistently high results were returned from Lion's Ear where four of the five samples collected returned TREO grades exceeding 1%, up to a maximum of 12.8% TREO, by far the highest assay returned from the programme.

These results identified a distinct variation between the Western Belt that was sampled between Yangibana North and Lion's Ear, and the southern prospects between Tongue, Yangibana, and Yangibana South. The higher TREO grades appear to occur in the north, but the higher (Nd_2O_3 Pr_6O_{11}):TREO ratios were found to occur in the southern prospects.

A second site visit was carried out by Hastings in April 2012 to locate and sample the remaining ironstone outcrops that had not been accessed during the site visit in September 2011. Prospects visited during this visit were Hook and Kane's Gossan. Hook South was not reached during the visit. As with the previous visit, a scintillometer was used to identify areas with the highest radiation values on the assumption that high radioactive content (Th-U) correlates with high REE content. Grab samples were collected during the visit and analysed by Genalysis Perth. The best results returned were 3.58% TREO including 0.85% Nd_2O_3 Pr_6O_{11} at Hook, and 1.96% TREO including 0.50% Nd_2O_3 Pr_6O_{11} at Kane's Gossan.

During 2014 Hastings completed two drilling programs. The first of these programmes was completed during May 2014, and consisted of 1,836m of RC drilling across 44 drillholes. All drillholes completed during this programme were located at the Yangibana North prospect. These drillholes were completed across 9 lines in a northeasterly orientation, with drillholes completed at a nominal 50 metre spacing. Drillholes ranged in depth from 12 metres to 90 metres.

All holes completed during this programme intersected the target ironstone/quartz unit, surrounded by an alteration halo of fenitised granite of variable thickness. The ironstone unit was found to be flat-dipping toward the south and remained open both along strike and at depth.

Following the success of the RC drilling completed during May 2014, a second drilling programme was completed between August and October 2014. This drilling was completed across 8 distinct prospects within the Yangibana Project, comprising Yangibana North, Gossan, Lion's Ear, Hook, Kane's Gossan, Bald Hill North, Bald Hill South, and Fraser's. Drilling had been planned for Yangibana South but was not completed. This drilling programme consisted of 6,582 metres of RC drilling, and 171.45 metres of diamond drilling. A total of 116 drillholes were completed comprising 107 RC drillholes, 6 RC pre-collars with diamond tails, and 3 diamond drillholes from surface.

A total of 70 rock chip samples were collected from regional targets during May 2014. Of these 70 samples, 36 were collected from carbonatite sills, ironstone units, and fenitic alteration zones in the southwest of the project area, where only limited work has previously been completed. The aim of this sampling was to confirm that the carbonatite sills and fenitic alteration zones carried anomalous REE mineralisation, and whether the ironstone outcrops in the area warranted further investigation. Results from these samples indicated that the carbonatite sills were anomalous in REE, however due to the generally narrow widths, and low grade, are not currently considered priority targets for further exploration in the short term.

32 samples were collected in the Bald Hill area, 28 of which were collected from the main Bald Hill ironstone outcrop. Of this 28 samples, 19 returned assays of above 0.5% TREO, with a maximum of 6.06% TREO including 2.71% Nd_2O_3. Six of these samples also returned greater than 5,000 ppm Nb, with a maximum of 66,010 ppm Nb_2O_5.

Digital aerial photography and DTM surveying was completed in during late 2014. This resulted in high resolution aerial photography across the target prospects within the Yangibana Project, the generation of a detailed DTM to be utilised in resource estimation.

Phase 3 Drilling was completed between May-August 2015 and consisted of 268 drillholes. This comprised 248 RC drillholes, and 20 diamond drillholes. Drilling focused on Bald Hill South, Fraser's, and Yangibana West. Drilling at Fraser's and Bald Hill South was focused on extending mineralisation identified during Phase 2 drilling along strike, and at depth. Drilling at Yangibana West focused on testing mineralisation directly along strike to the northwest of Yangibana North.

Additional work completed during 2015 included updated resource estimation, the collection of 22 rock chip samples, geological mapping at Bald Hill and Fraser's, petrographic analysis of six diamond core samples, ongoing metallurgical testwork, and various geotechnical, hydrogeological, environmental, and financial studies forming part of a Pre-Feasibility Study for the Yangibana Project.

Two field visits were completed in December 2015 and January 2016, with the aim of sampling a number of new target areas. These field visits focused on and area to the southwest of Fraser's where GSWA mapping indicated a number of ironstone veins. Rock chip sampling in this area retuned up to 1.90% (Nd_2O_3 Pr_6O_{11}) and 3.67% TREO. Importantly, the rock chip samples in this area indicated that the ironstone units have a high (Nd_2O_3 Pr_6O_{11})/TREO ratio, similar to the higher value deposits at Bald Hill and Fraser's. The most promising ironstone units identified from this sampling were named Auer, Mosander, Demarcay, Leceq, and Hisinger and prioritised for drill testing.

During 2016, 305 drillholes were completed for a total of 15,090.29m. Drilling was focused on Bald Hill and Fraser's and was designed as infill drilling with the aim of upgrading the previous Indicated resource areas to Measured category. A number of new targets were also drill tested, which had been identified through the earlier geochemical sampling, and available geophysics. These new targets comprised Auer, Auer North, Mosander, Demarcay, and Hatchett. Updated resource estimation was subsequently completed following the 2016 drilling programme.

Minor diamond drilling for geotechnical purposes was also completed over areas proposed for tailings storage facilities, and the processing plant within E09/2007 and E09/2018 respectively, as part of the ongoing Definitive Feasibility Study (DFS).

During June and July of 2016, high resolution airborne magnetic and radiometric data was acquired over the Yangibana Project (Registration ID R71264). Acquisition was completed by Thomson Aviation, with 17,908 line kilometres of data acquired at a line spacing of 25m. Processing and interpretation was completed by Southern Geoscience Consultants (SGC). SGC identified 108 thorium anomalies within the radiometric dataset, with 22 areas identified as priorities for further exploration.

Digital aerial photography and DTM surveying was also completed during 2016 by HyVista Corporation, to extend the existing survey area as completed in 2014 (Registration ID R71266). The data acquired in 2016 was merged with the 2014 survey, to provide DTM and airphoto coverage over new target areas, and areas of proposed mining infrastructure.

During 2017 a total of 451 drillholes were completed on the Yangibana Project, for a total of 30,025.94m, including 3105.94m of diamond drilling (Figure 5). Drilling included resource definition and extension drilling at Bald Hill, Fraser's Yangibana North, Yangibana West, Auer, and Auer North, exploration drilling at Terry's Find, Yangibana, and Simon's Find, and a programme of water exploration drilling. This drilling resulted in updates to Resource estimations for areas with existing Resources, and initial resource estimation for Yangibana and Simon's Find.

Further rock chip sampling, XRD mineralogical analysis, and petrological analysis were also completed during 2017.

Figure 5 shows all drilling completed within the Yangibana Project to date.

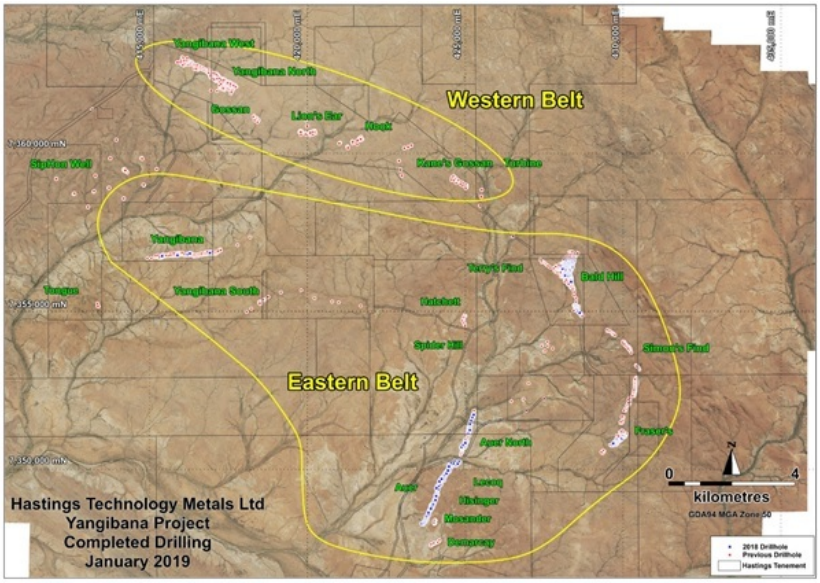


Figure 5: Yangibana Project - Completed Drilling 2018 shown in blue

7. Current Exploration

Exploration completed during 2018 consisted of drilling, updated resource estimation, some minor geochemical sampling and some local geological mapping.

Table 1: Summary of Exploration Completed Within C265/2008

Tenement	Drillholes	Geochemical Samples	Metallurgical Testwork	Resource Estimation	Petrology	Geotechnical	Geophysical
E09/1043	-	-	-	-	-	-	-
E09/1700	28	-	X	X	X	X	X
E09/1703	-	-	-	-	-	-	-
E09/1704	-	-	-	-	-	-	-
E09/1705	-	-	-	-	-	-	-
E09/1706	-	-	-	-	-	-	-
E09/1943	-	5	-	-	X	-	-
E09/1944	-	12	-	-	-	-	-
E09/1989	162	10	X	X	X	X	X
E09/2007	-	9	-	-	X	-	-
E09/2018	-	6	-	-	-	-	X
E09/2084	-	-	-	-	-	-	-
E09/2086	-	-	-	-	-	-	-
E09/2095	-	-	-	-	-	-	-
E09/2129	-	-	-	-	-	-	-
E09/2137	-	-	-	-	-	-	-
P09/482	-	-	-	-	-	-	-
P09/489	-	-	-	-	-	-	-
M09/157	131	-	X	X	-	-	-
M09/158	42	-	X	X	-	-	X
M09/159	-	5	-	-	-	-	-
M09/160	-	-	-	-	-	-	-
M09/161	-	-	-	-	-	-	-
M09/162	-	-	-	-	-	-	-
M09/163	-	7	-	-	-	-	-
M09/164		-	-	-	-	-	-
M09/165	9	-	X	-	-	X	-

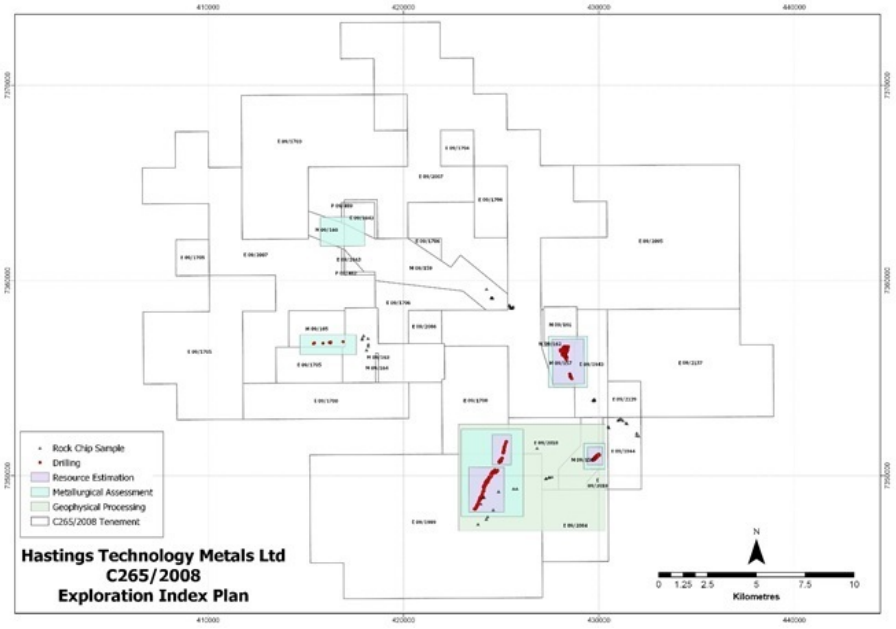


Figure 1: Exploration Index Plan

8. Current Exploration Summary

8.1 Data Review

Throughout 2018 the Hastings exploration team continued to review existing data across all tenements within the project area in order to identify further targets that require assessment. Based on the Company's project wide datasets including hyperspectral, magnetic, radiometric, and aerial photography, various areas that have been subject to limited work to date have been identified as requiring further field assessment. Table 7 summaries the completed review on a tenement by tenement basis.

TENEMENT	PROSPECTS	PLANNED ACTIVITY FOLLOWING REVIEW
100% HASTINGS		
M09/157	Bald Hill	Metallurgy; Mine Planning; Infill and Extension Drilling
M09/158	Fraser's	Metallurgy; Mine Planning; Infill and Extension Drilling
M09/160	Yangibana West	Metallurgy; Mine Planning; Infill and Extension Drilling
M09/162	Bald Hill	Metallurgy; Mine Planning; Infill and Extension Drilling
M09/164	Yangibana South	Increase Sample Density on Ironstone Outcrop; Exploration Drilling
M09/165	Yangibana	Increase Sample Density and Mapping on Western Extension of Yangibana; Metallurgy; Mine Planning; Infill and Extension Drilling
P09/482	Yangibana North Deeps	Review Data and Assess Potential for Depth Extensions to Yangibana North; Drill Test
P09/489	Yangibana North	Review Available Data to Determine Prospectivity
E09/1700	Auer North; Spider Hill; Hatchett	Auer North Metallurgy; Auer North Mine Planning; Auer North Extension and Infill Drilling; Increase Mapping and Sampling Coverage of Hatchett and Spider Hill and Assess Niobium Potential
E09/1943	Simon's Find	Increase Mapping and Sampling Coverage of Simon's Find; Infill and Extension Drilling
E09/1944	Simon's Find	Increase Mapping and Sampling Coverage of Simon's Find; Infill and Extension Drilling
E09/1989	Fraser's Southwest	Fraser's Southwest, mapping and sampling
E09/2007	SipHon Well	Water Exploration in Palaeochannel Areas; Mapping and Sampling Over Various Regional Targets
E09/2018	Fraser's Southwest	Drill Testing of T1 Magnetic Target
E09/2084	Fraser's South	Mapping and Sampling of Extensions to Ironstoe/Carbonatite Features Evident from Geophysical and Aerial Imagery
E09/2086	Central Area	Mapping and Sampling and Basic Reconaissance over Features Identified in Hyperspectral and Geophysical Data
E09/2095	Regional	More Detailed Field Reconnaissance of Targets Identified in Geophysical Data
E09/2129	Simon's Find	Increase Mapping and Sampling Coverage of Simon's Find; Infill and Extension Drilling
E09/2137	Regional	More Detailed Field Reconnaissance of Targets Identified in Geophysical Data
70% HASTINGS		
M09/0159	Yangibana North - Kane's Gossan	Infill and Extension Drilling; Increase Mapping and Sampling Coverage Across Tenement
M09/0161	Bald Hill North	Detailed Mapping and Sampling
M09/0163	Yangibana South; Yangibana East	Increase Sample Density and Mapping on Yangibana South and Yangibana East Ironstone Outcrops and Subsidiary Structures; Exploration Drilling
E09/1043	Yangibana North Deeps	Review Data and Assess Potential for Depth Extensions to Yangibana North; Drill Test
E09/1703	Yangibana West extension	Review and Process Geophysical Data for Yangibana West Western Extension to Develop Drill Targets.
E09/1704	Regional	Field Reconnaissance
E09/1705	SipHon Well; Tongue	Water Exploration in Palaeochannel Areas; Increase Mapping and Sampling Coverage of Tongue
E09/1706	Central Area	Field Reconnaissance

Table 7: Summary of Review of Targets for All Tenements

8.2 Geological Mapping

ArcGIS software has now been utilised to begin mapping of the local geology. Primary focus is on the ironstones in the 100% Hastings tenements. Mapping in previosuly drilled prospects is being used to further pinpoint any future drilling in these areas. Figure 4 shows a minimized version of the map completed to date. The primary outcrops can be seen as dark dashes on the map.

8.3 Surface Sampling

GEOCHEMICAL SAMPLING
A total of 43 rock chip samples were collected during the reporting period across the Yangibana Project (Figure 4). The bulk of this sampling was completed over specific targets identified through previously completed exploration. The samples were taken primarily from Kane's Gossan, Demarcay, Leceq and Yangibana.
Sampling carried out in Kane's Gossan, Demarcay and Leceq was to assist in more accurately pinpointing any future drilling in these areas, and to show continuity of mineralisation along strike.
At Yangibana the sampling was carried out to cover previously untested ironstone units as shown on the Edmund 1:100,000 map sheet to determine their prospectivity.
A series of samples were also collected off the eastern margin of the Gifford Creek Ferro-Carbonatite Complex, within the Irregully Formation of the Edmund Group. These samples were collected to determine whether a potential source of carbonate could be located in this area for use during processing of the REE ore. Samples, however, were predominantly weathered and bleached silicic siltstones to fine sandstone and did not indicate the presence of the mapped carbonates.

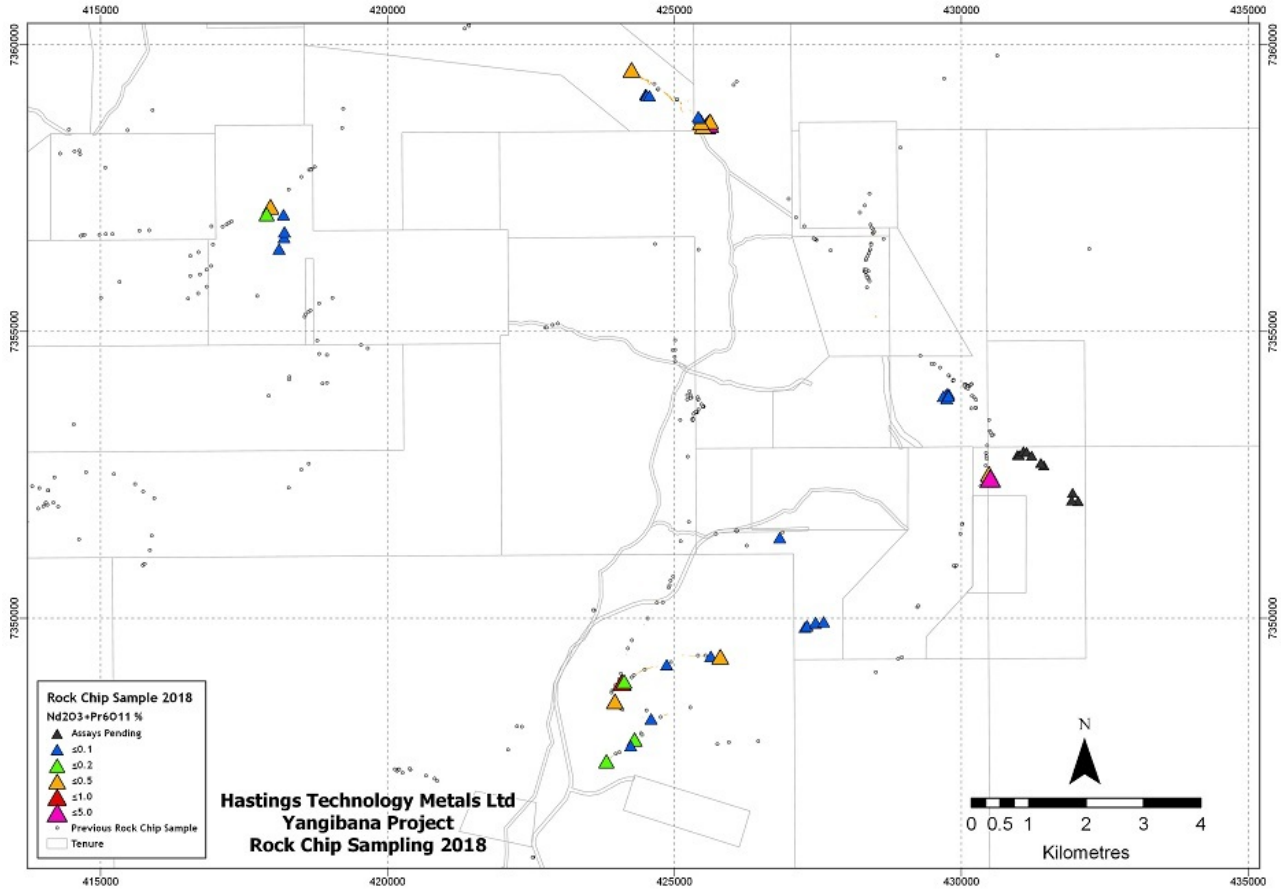


Figure 4: Geochemical Sampling 2018

8.4 Drilling

During 2018 a total of 372 drillholes were completed on the Yangibana Project, for a total of 20,126.05m, including 2969m of diamond drilling (Figure 5).
Drilling was focussed on metallurgical sampling at Bald Hill and Fraser's and on resource definition at Auer and Auer North.
Geotechnical drilling was carried out in Yangibana, Auer and Auer North. 19 geotechnical holes were drilled for a total of 1887.4m which is included in the total metres above.

Bald Hill and Bald Hill Southeast
A total of 131 drillholes were completed at Bald Hill and Bald Hill Southeast during 2018, including 5 diamond drillholes, for a total of 4,268.05m (Figure 6).
Drill holes at Bald Hill were mostly collared 10m from previous holes in order to commence the grade control process while also providing the required representative metallurgical sample. The DD holes in Bald Hill were for comminution sampling.
Future drilling in Bald hill will be carried out on the Southwest of the prospect where there is evidence the ore body may continue at depth.

Table 2: Summary of Bald Hill and Bald Hill Southeast 2018 best Intersections

Hole ID	Depth From	Depth To	Interval	TREO (%)	Nd_2O_3 Pr_6O_{11} %
BHRC337	24	34	10	1.74	0.71
BHRC338	33	40	7	2.36	0.95
BHRC342	8	15	7	2.20	0.96
BHRC399	34	48	14	1.48	0.59
BHRC425	3	24	21	1.64	0.63
BHRC437	0	16	16	1.29	0.55
BHRC443	2	16	14	1.78	0.77

Fraser's
A total of 42 drillholes were completed at Fraser's, including 3 diamond drillholes, for a total of 2,109.5m (Figure 8). Results from the infill reverse circulation (RC) drilling at the Fraser's deposit are in line with expectations.
The drilling in Fraser's was also primarily for metallurgical purposes with the diamond drillholes being for comminution testwork.

Table 3: Summary of Fraser's 2018 Best Intersections

Hole ID	Depth From	Depth To	Interval	TREO (%)	Nd_2O_3 Pr_6O_{11} %
FRRC136	23	32	9	2.38	1.01
FRRC153	94	98	4	2.80	1.15
FRRC165	11	20	9	3.03	1.22
FRRC168	30	36	6	2.87	1.16

Yangibana
A total of 9 drillholes were completed at Yangibana, all of which were Diamond holes, for a total of 623.9m (Figure 9). 5 of the holes drilled in Yangibana were geotechnical holes with the remaining 4 holes being for comminution purposes.

Auer and Auer North
A total of 190 drillholes were completed at Auer and Auer North, including 24 diamond drillholes, for a total of 13,124.6m, 1,977.6m of this was DD (Figure 10; Figure 11). Drilling in Auer and Auer North was primarily for resource definition purposes though some drilling carried out to the South extended past previous exploration.
Of the 24 Diamond holes previously mentioned in Auer and Auer North, 14 were for geotechnical purposes. The geotechnical holes comprised 1417.5m of the metres above. The remaining 10 diamond holes were for comminution purposes.

Table 4: Summary of Auer and Auer North 2018 Best Intersections

Hole ID	Depth From	Depth To	Interval	TREO (%)	Nd_2O_3 Pr_6O_{11} %
AURC176	40	53	13	1.60	0.49
AURC180	121	128	7	2.57	0.85
AURC186	17	25	8	2.04	0.84
AURC121	98	120	22	0.96	0.39
AURC137	10	12	2	4.02	1.38
AURC202	85	94	9	1.85	0.65
ANRC084	63	73	10	1.87	0.60
ANRC093	53	56	3	2.24	0.72

8.5 Resource Estimations/Revisions

Following completion of the 2018 drilling programme, resource estimation was completed. Resource estimation was completed by Widenbar and Associates Pty Ltd, resulting in a global resource for the Yangibana Project of 21.67 Mt at 1.17% TREO including 0.40% Nd_2O_3 + Pr_6O_{11} (Measured, Indicated, and Inferred). The resource estimate broken down by resource category is provided in Table 5. Resource tonnages by deposit are summarised in Table 6. Full details of the resource estimation are provided in '*Yangibana Project Mineral Resource Estimate October 2018*' provided as an appendix.

Category	Tonnes	Nd_2O_3+$Pr6O_{11}$	TREO	Nd_2O_3	Pr_6O_{11}	Dy_2O_3	Tb_4O_7
		%	%	ppm	ppm	ppm	ppm
Measured	4,700,000	0.42	1.17	3,394	848	59	19
Indicated	8,700,000	0.41	1.24	3,278	843	46	16
Meas+Ind	**13,400,000**	**0.42**	**1.22**	**3,319**	**845**	**50**	**17**
Inferred	8,300,000	0.36	1.09	2,836	737	41	13
TOTAL	**21,700,000**	**0.39**	**1.17**	**3,134**	**804**	**47**	**15**
Note: Rounding errors may occur							

Table 5: 2018 Global Mineral Resource Estimate - Yangibana ProjectTable 5: 2018 Global Mineral Resou

Deposit	Tonnes	TREO	Nd_2O_3+$Pr6O_{11}$	Nd_2O_3	Pr_6O_{11}	Dy_2O_3	Tb_4O_7
		%	%	ppm	ppm	ppm	ppm
Bald Hill	5,710,000	1.01	0.40	1.01	3,205	765	56.76
Bald Hill SE	540,000	0.65	0.27	0.65	2,172	499	55.35
Fraser's Main	1,060,000	1.39	0.59	1.39	4,718	1,206	62.60
Fraser's North	330,000	0.51	0.23	0.51	1,885	446	30.45
Fraser's South	80,000	0.89	0.27	0.89	2,087	649	20.09
Yangibana	2,170,000	0.84	0.40	0.84	3,397	594	34.98
Yangibana West	2,540,000	1.29	0.35	1.29	2,674	779	44.21
Yangibana North	3,430,000	1.79	0.46	1.79	3,500	1,080	44.13
Auer North	680,000	1.03	0.35	1.03	2,677	788	21.79
Auer	2,000,000	1.09	0.38	1.09	3,004	797	52.29
Lion's Ear	710,000	1.54	0.39	1.54	3,003	911	48.48
Kanes's Gossan	570,000	1.04	0.29	1.04	2,182	680	38.90
Hook	290,000	1.52	0.33	1.52	2,479	855	28.86
Gossan	250,000	1.43	0.35	1.43	2,658	857	23.00
Simon's Find	1,310,000	0.66	0.35	0.66	2,951	528	40.66
TOTAL	21,670,000	1.17	0.39	1.17	3,132	803	46.83

Table 6: Resource Tonnage and Grade by Deposit - 2018 Mineral Resource Estimate

8.6 Petrology

In the first petrographic report 24 rocks from two prospects, Spider Hill (7 samples) and Simon's Find (17 samples), were examined. A 1.25hr microprobe session at UTAS Central Science Laboratory was carried to confirm the composition of the extremely fine-grained phases in a number of ironstone samples which could not be confidently identified petrographically.
The samples from Spider Hill included 3 veined, medium-grained monzonites with well developed 'ironstone' veining, a foliated syenite to microsyenite, a contact between a micromonzonite and an aplite, two entirely metasomatic rocks and a banded fault zone rock composed almost entirely of smashed vein quartz and Fe oxy-hydroxides. The monzonites have K-spar megacrysts or coarse phenocrysts up to 1cm long, and all three monzonites show a heavily fractured fabric, with seams and veinlets of pale brown phlogopite and scattered magnetite euhedra apparently invading along fractures and grains boundaries.
Samples from Simon's find ranged from heavily fractured quartz veins with either colourless clays or red Fe oxy-hydroxides filling voids and openings along fractures to very even-textured, strongly weathered and altered ironstone composed almost entirely of fine-grained quartz and magnetite that is now almost entirely replaced by hematite and Fe oxy-hydroxides.
A total of 14 samples from drillholes at Auer were submitted to A & A Crawford Geological Research Consultants for petrographical analysis. The samples examined were obtained from holes AUDD178, AUDD182 and AURC198. The rocks from AUDD178 and AUDD182 are all phoscorite-syenite association rocks. The RC chips in 5 slides prepared from the three 1m intervals between 49m and 52 m in hole AURC198 were found to contain only regolithic Fe oxy-hydroxide material, vein quartz and weathered siliceous banded Fe formation metasediments, with no chips from phoscorite-syenite association rocks; these were of no further interest.
The reports provided by A & A Crawford Geological Research Consultants are provided as appendices.

8.7 XRD Analysis

Utilisation of the Company's pXRD equipment has continued to prove invaluable in determining relationships between minerals associated with the ore body. Adding to its value in distinguishing between the ironstone hosted mineralisation and the siderite hosted mineralisation, a new theory has formed regarding the relationship between the monazite and the presence of Al-phosphates. It is now believed some of the Al-phosphates are present as a result of alteration and hydration of the monazite in some clay rich samples.
The primary focus this year was on samples from Auer and Auer North with slightly more emphasis being put on the halo of the mineralisation.
Use of the pXRD will continue in 2019 with hopes of gathering more evidence regarding the pre-mentioned theory and discovering new mineral relationships involved with the ore body.

8.8 Metallurgy

Metallurgy was one of the primary reasons for the bulk of the drilling completed in Bald Hill and Fraser's with the samples collected from the RC drilling being used to test how the processing plant will cope with the complexity of the ore body. The beneficiation process was carried out at ALS metallurgy's commercial lab in Balcatta where 19 tonnes of ore collected from this drilling programme was processed. The plant was operating for 24 hours a day for 8 days to process the ore.
Comminution samples were also taken from all diamond holes completed in Bald Hill and Fraser's, as well as from all non-geotechnical diamond holes in Yangibana, Auer and Auer North.
Further metallurgical tests will have to be carried out on samples from Auer and Auer North to determine variability.

8.9 Geophysical Processing

During 2018 Southern Geoscience Consultants was engaged to perform further processing of airborne magnetic data acquired during 2016.

This processing was focused on six target areas proximal to the Auer and Auer North deposits. This processing utilised a combination of inversion modelling, forward modelling, and depth filtering to better define magnetic targets.
The inversion modelling was successful in modelling a feature with a strong magnetic susceptibility over a strike length of approximately 1 kilometre, with a depth extent of 800m (Figure 12; Figure 13). An application for co-funding through the Exploration Incentive Scheme was made for drill testing of this target (T1) and subsequently granted, with drilling planned for 2019.
Forward modelling was used to provide modelled conductors for a series of discrete linear magnetic features. The most significant of these were the strike extensions to Auer to the south, and Auer North to the north, where outcrop is lost below cover. Modelling identified tabular bodies broadly consistent with the Auer and Auer North orebodies in both directions. Drill testing of the southern strike extension of Auer was completed during 2018, successfully intersecting mineralisation (Figure 10).
Depth filtering was also implemented to enhance a series of magnetic features in the area, with further follow up work required.
The report provided by Southern Geoscience Consultants is provided as an Appendix.

9. Conclusion and Recommendations

Infill drilling completed during 2018 was successful in converting resources into reserves in some of the primary prospects, Bald Hill, Fraser's, Auer and Auer North. The drill programme was also successful in generating the amount of metallurgical sample necessary to carry out the test with the pilot plant.
A minor increase to resources was achieved due to an extension of the drilling to the south of Auer.
The geotechnical drilling programme with holes in Auer, Auer North and Yangibana was completed successfully.
Further drilling is planned for 2019 with the first 2 holes of the programme testing an aeromagnetic target with a strike length of 1km and a strong magnetic susceptibility. These holes will far exceed the depth of any previous holes in the area.

10. References

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11. Appendices

No Appendices as text are available